

Topic 03: Existence and Uniqueness of Solutions (Picard-Lindelöf)

Professor Leonard Master Series · Lesson 11

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Two Big Questions

When an engineer writes down a differential equation for an airplane wing or a chemical reactor, they must know:

1. **Does a physical solution actually exist?** (Will the reactor blow up because math has no answer?)
2. **Is the solution unique?** (If I run the experiment twice under identical initial conditions, will I get the exact same behavior?)

Professor Leonard explains:

"Picard's Theorem is like an insurance policy. It gives you conditions on your equation that guarantee you will get one and only one trajectory through your initial condition point."

2. Picard-Lindelöf Theorem Protocol

Given the Initial-Value Problem:

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

The Two Tests:

1. **Existence Test:** Is $f(x, y)$ continuous in a rectangle around (x_0, y_0) ?
 - If **YES**, at least one solution exists!
 2. **Uniqueness Test:** Is the partial derivative $\frac{\partial f}{\partial y}$ continuous in that same rectangle?
 - If **YES**, that solution is **strictly UNIQUE!** No other curve can pass through (x_0, y_0) .
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3. The Classic Exam Trap: Branching Solutions

Problem: Examine $\frac{dy}{dx} = \frac{x}{y}$, with initial condition $y(0) = 0$.

- **Step 1: Inspect $f(x, y)$:**

$$f(x, y) = \frac{x}{y}$$

At $(0, 0)$, $y = 0$, which causes division by zero! $f(x, y)$ is **discontinuous** at the origin.

- **Step 2: Physical Consequence:** Separating variables: $ydy = xdx \implies \frac{1}{2}y^2 = \frac{1}{2}x^2 + C \implies y^2 - x^2 = 2C$. For $y(0) = 0, C = 0 \implies y^2 = x^2 \implies y = x$ or $y = -x$. Notice two distinct straight lines pass through $(0, 0)$! Uniqueness completely fails because the continuity condition was violated.
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4. Leonard's Golden Rules

- ★ **Always compute $\frac{\partial f}{\partial y}$ immediately:** Do not try to solve the ODE first. Compute the derivative with respect to y , look at the denominator, and find where it equals zero. If your initial point (x_0, y_0) makes that denominator zero, uniqueness is not guaranteed!